pygeotypes: A Dependency-Light Shape-Catalogue Library with Class-Conditional Conformal Assignment for Physical Response Signals

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***Abstract*—Many physical systems are characterised through the *shape* of a response signal: the pressure transient of a fractured reservoir, the drawdown of a hydrogeology pumping test, the temperature curve of a thermal-response test. A recent methodology, GeoTypes (Kamel Targhi *et al.*, 2026), turns an ensemble of such signals into a catalogue of behaviour types by clustering their shapes. Two practical gaps stand between that methodology and routine use. First, the natural building blocks in mid-2026 are awkward to deploy: the maintained k-medoids implementations are either unmaintained (scikit-learn-extra), copyleft (the fast Rust `kmedoids`, GPL-3), or drag native dependencies that do not run in a browser sandbox (tslearn, aeon). Second, assigning a new signal to a catalogue by nearest medoid gives no guarantee and no way to say "this shape is unlike anything in the catalogue". We present `pygeotypes`, a dependency-light library whose core is pure NumPy/SciPy, so it runs unchanged offline and in the browser (Pyodide), that packages the full GeoType pipeline (log-resampling, Bourdet derivative, band-constrained dynamic time warping, PAM k-medoids with silhouette model selection, an exact-JSON catalogue artifact) and adds a *class-conditional split-conformal assignment layer*. The conformal layer returns, for a new signal, a per-type p-value and the prediction set of behaviour types consistent with it at a requested confidence, and an empty set is an honest out-of-catalogue flag. On a reproducible pressure-transient benchmark the empirical coverage tracks or exceeds the nominal $1-\alpha$ target across the operating range (0.960 at α=0.05, 0.938 at 0.10, 0.885 at 0.15, 0.800 at 0.20), the prediction sets stay informative (mean size 1.0 down to 0.7), and every alien shape tested is flagged out-of-catalogue at all confidence levels. We describe the library, the conformal construction, and the validation; every number and figure regenerates from the committed artifacts.**

***Index Terms*—Shape clustering, dynamic time warping, k-medoids, conformal prediction, out-of-distribution detection, pressure-transient analysis, well testing, reproducible software, Pyodide.**

I. Introduction

A large family of characterisation problems reduces to reading the *shape* of a response signal. In well testing, the Bourdet derivative [7] of a pressure transient traces the flow regimes of the reservoir, and a dual-porosity system leaves a characteristic valley between two radial plateaus [8]. In hydrogeology a pumping-test drawdown, and in geothermics a thermal-response temperature curve, carry the same kind of diagnostic shape. Kamel Targhi *et al.* [1] formalised this into a methodology they call GeoTypes: take an ensemble of response signals, compare them by shape, cluster them, and read off a small catalogue of behaviour types whose medoids are interpretable prototypes. The methodology is theirs; this note is about making it *reusable* and *decision-ready* as software.

Two gaps stand in the way. The first is deployment. The shape comparison wants dynamic time warping [4] and the clustering wants PAM k-medoids [5] with silhouette model selection [6], but in mid-2026 the convenient implementations are hard to ship: scikit-learn-extra (k-medoids) is unmaintained, the fast Rust `kmedoids` is GPL-3 (a licensing constraint for a permissive library), and tslearn and aeon pull in Numba or native code that does not run in a Pyodide browser sandbox. A characterisation tool that a practitioner can run from a notebook, or that a static web application can run client-side, needs a dependency-light core. The second gap is trust. Assigning a new signal to the catalogue by its nearest medoid always returns *some* type, even for a signal whose shape belongs to no catalogued behaviour; it offers neither a confidence statement nor an out-of-catalogue flag, which is exactly what a characterisation workflow must surface.

`pygeotypes` closes both gaps. Its core is pure NumPy/SciPy, so the identical code runs offline and in the browser; the accelerated and attribution features are optional extras. It packages the whole GeoType pipeline behind a small API with an exact-round-trip JSON catalogue artifact, and, as the one methodological addition of this note, it wraps assignment in a *class-conditional split-conformal* layer that turns the catalogue into a decision tool with finite-sample, per-class coverage guarantees [2], [3] and an honest out-of-catalogue flag. The contributions of this software note are therefore: (i) a reusable, browser-ready implementation of the GeoType methodology; and (ii) the conformal assignment layer, validated empirically. We are explicit that the underlying shape-catalogue methodology is the contribution of Kamel Targhi *et al.* [1], cited wherever it is used.

II. Statement of Need

The users are practitioners and researchers who characterise systems from response-signal shape and who need the GeoType methodology to be (a) installable and portable and (b) usable as a decision tool. A reservoir or hydrogeology analyst

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Software note, 2026. Package `pygeotypes` on PyPI (https://pypi.org/project/pygeotypes); source (MIT) at https://github.com/fsantibanezleal/CAOS_GeoTypes. This note describes a reusable software implementation of the GeoType methodology of Kamel Targhi *et al.* (2026) and adds a conformal assignment layer; the methodology itself is their contribution, cited throughout.

wants to drop a new pressure or drawdown curve against an established catalogue and be told, with a confidence level, which behaviour types are consistent with it, or that none are. A static web application built for such a workflow must run the assignment client-side, which rules out native dependencies. And any honest characterisation must be able to say "out of catalogue" rather than silently snapping an unusual signal to the nearest prototype. No single existing package covers this combination: the maintained clustering stacks are either unshippable under a permissive license or non-portable to the browser, and none attaches a distribution-free guarantee to catalogue assignment. `pygeotypes` is built to fill exactly that combination, with the methodology of [1] as its scientific basis.

III. The Pipeline (Packaged Methodology)

The pipeline (Fig. 1) follows [1] and is implemented as five small modules. The **preprocess** stage resamples each raw (t, p) signal onto a common logarithmic time grid, computes the Bourdet pressure derivative [7] (optionally the second log-derivative, which is offset-free), and normalizes, so that only shape, not absolute level, drives the comparison. The **distance** stage computes a Sakoe-Chiba band-constrained dynamic time warping distance [4] between curves (a pure-NumPy dynamic program, with an optional compiled backend offline that is parity-tested against it); the band permits modest regime-timing shifts while forbidding degenerate alignments. The **cluster** stage runs PAM k-medoids [5] on the precomputed distance matrix (BUILD then SWAP, multi-start, seeded) and selects the number of types by the silhouette [6]. The **catalogue** stage packages the medoid curves, labels, preprocessing parameters and provenance into a `Catalogue` artifact with an exact JSON round-trip, so a catalogue baked offline is replayed byte-for-byte in a browser. Figure 2 shows a catalogue the library builds from a synthetic dual-porosity ensemble: two interpretable behaviour prototypes that differ in the timing of the interporosity transition. An optional **attribute** module (the only feature needing scikit-learn) relates catalogue membership back to physical drivers with a Spearman-pruned random forest cross-checked by TreeSHAP [10] and permutation importance, but it is outside this note's scope.

IV. Class-Conditional Conformal Assignment

The methodological addition is the assignment layer. Nearest-medoid assignment returns $\arg\min_g d(x, \mathrm{medoid}_g)$ for a new curve x, always a type and never a caveat. Split-conformal prediction [2], [3] turns this into a guaranteed decision. Fit on a *calibration* set of labelled curves disjoint from those used to build the catalogue. For each type g the calibration scores are the DTW distances of the class-g calibration curves to medoid g, $S_g = \{\, d(x_i, \mathrm{medoid}_g) : y_i = g \,\}$ (the nonconformity score is the same DTW distance the catalogue is built on, so the layer stays pure NumPy and browser-safe). For a new curve x the conformal p-value for type g is the smoothed rank of its distance among the class-g calibration scores,

$$p_g(x) \;=\; \frac{\big|\{\, s \in S_g : s \geq d(x, \mathrm{medoid}_g) \,\}\big| + 1}{|S_g| + 1}, \tag{1}$$

and the prediction set is every type whose p-value clears the level,

$$\Gamma^{\alpha}(x) \;=\; \{\, g : p_g(x) > \alpha \,\}. \tag{2}$$

Because the calibration is done *per class* (a Mondrian taxonomy over the labels), the guarantee is class-conditional: under exchangeability of the class-g calibration and test curves, the true type is in the set with probability at least $1 - \alpha$ *within each class*,

$$\Pr\big[\, g \in \Gamma^{\alpha}(x) \,\big|\, y = g \,\big] \;\geq\; 1 - \alpha. \tag{3}$$

Class-conditional validity matters because GeoType populations are imbalanced in real ensembles (Fig. 2), and a merely marginal guarantee could under-cover a minority behaviour. Two outputs follow naturally. A *prediction set* larger than one signals genuine ambiguity between behaviours at the requested confidence; a singleton is a confident call. An *empty set* is an honest out-of-catalogue flag: the curve's shape is consistent with no catalogued behaviour at confidence $1 - \alpha$, exactly the case a characterisation workflow must surface instead of forcing the nearest medoid. The calibration scores are small sorted arrays baked into the catalogue artifact, so the browser recomputes p-values and sets live.

V. Empirical Validation

We validate the conformal layer with a reproducible experiment (Fig. 3) on synthetic pressure transients so the ground truth is controlled. An ensemble of Warren-Root dual-porosity responses [8] (generated by Stehfest inversion [9] of the analytical solution over a wide parameter span, giving two well-separated behaviour modes) is preprocessed, clustered into a $K{=}2$ catalogue, and split into disjoint training (catalogue), calibration (320 curves) and test (400 curves) sets. The ground-truth type of a test curve is its nearest medoid under the same DTW metric, the self-consistent label for an unsupervised catalogue.

Three results follow. First, *the coverage guarantee holds empirically*: across the operating range the measured marginal coverage tracks or exceeds the nominal target at every level (0.960 vs 0.95; 0.938 vs 0.90; 0.885 vs 0.85; 0.800 vs 0.80; 0.738 vs 0.70), and the per-class coverages confirm it holds for both the majority (GeoType 0) and the minority (GeoType 1) behaviour, the property the class-conditional calibration is there to protect. Second, *the sets are informative*: mean prediction-set size falls from 0.99 at $\alpha{=}0.05$ to 0.74 at $\alpha{=}0.30$, so at operating confidence the layer typically returns a single confident type, not the whole catalogue. Third, *out-of-catalogue detection works*: every shape alien to the catalogue (high-frequency oscillations, unlike any pressure transient) is flagged out-of-catalogue at all tested confidence levels (α from 0.05 to 0.30), correctly refusing to force an alien signal onto the nearest medoid. The experiment is a single, seeded run; the numbers regenerate exactly from the committed script and artifact.

VI. Architecture and Availability

The library is five small pure-NumPy/SciPy modules (`preprocess`, `distance`, `cluster`, `catalogue`,

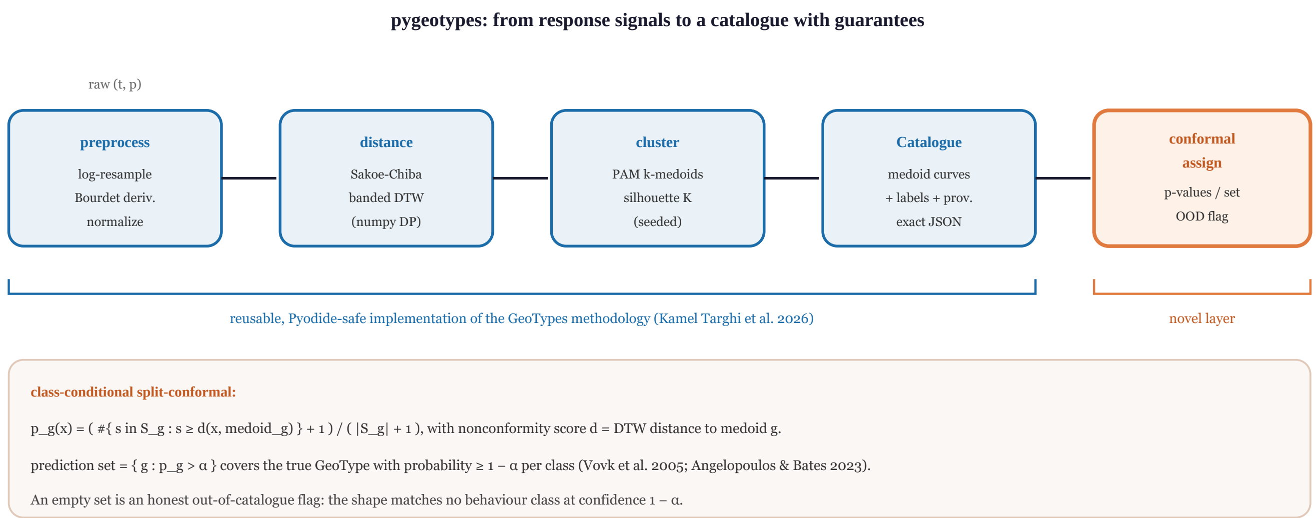

Fig. 1. The `pygeotypes` pipeline. The first four stages are a reusable, Pyodide-safe implementation of the GeoType methodology of Kamel Targhi *et al.* [1]: preprocess (log-resample, Bourdet derivative, normalize), band-constrained DTW distance, PAM k-medoids with silhouette model selection, and an exact-JSON `Catalogue` artifact. The final stage is the one methodological addition of this note: a class-conditional split-conformal assigner that returns per-type p-values, a prediction set, and an honest out-of-catalogue flag.

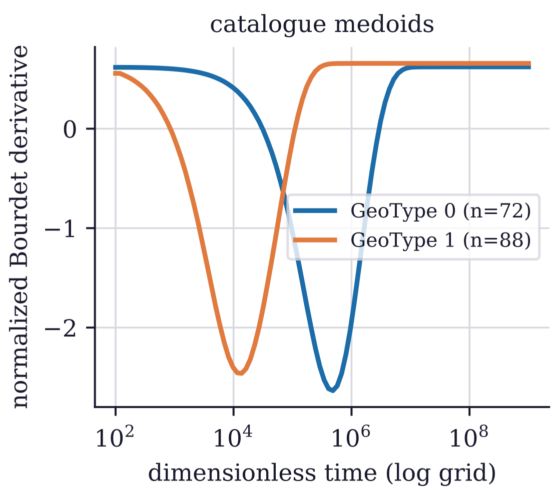

Fig. 2. A GeoType catalogue built by `pygeotypes` from a Warren-Root dual-porosity ensemble: two behaviour prototypes (medoids of the normalized Bourdet derivative on a common log grid) differing in the timing of the interporosity transition, an early-transition type and a late-transition type. Class sizes are imbalanced (72 and 88), which is why the conformal layer calibrates per class.

`assign`) plus the optional `attribute` module. Install lanes keep the core light:

```
pip install pygeotypes            # pure core (numpy
      + scipy)
pip install pygeotypes[fast]      # + dtaidistance (
     fast offline DTW)
pip install pygeotypes[attr]      # + scikit-learn +
      shap (drivers)
```

A catalogue is built and a new curve assigned in a few calls:

```
from pygeotypes import (dtw_matrix, select_k,
    pam_kmedoids,
    build_catalogue, ConformalAssigner)
D   = dtw_matrix(X, window=10)          # banded
    DTW
res = pam_kmedoids(D, k=select_k(D, range(2,9))["
    best_k"], seed=0)
cat = build_catalogue(X, t_grid, D, k=res.k,
    dtw_window=10, result=res)
asg = ConformalAssigner(cat).fit(X_cal, labels_cal
    )
out = asg.predict(new_curve, alpha=0.1)   # .
    prediction_set, .out_of_catalogue
```

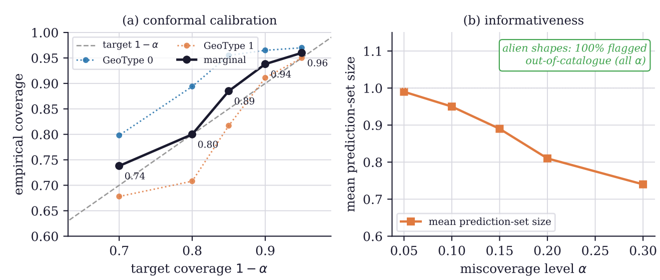

Fig. 3. Reproducible validation of the conformal layer on a Warren-Root pressure-transient benchmark (K=2 catalogue; 320 calibration, 400 test curves). **(a)** Empirical marginal coverage against the nominal $1-\alpha$ target, both marginally (bold) and per class: coverage sits on or above the diagonal across the operating range $(0.960, 0.938, 0.885, 0.800$ at $\alpha = 0.05, 0.10, 0.15, 0.20)$, and the class-conditional calibration keeps both the majority and the minority behaviour covered. **(b)** The prediction sets stay informative (mean size 0.99 down to 0.74), and every alien high-frequency shape is flagged out-of-catalogue at all tested confidence levels.

The core carries no native dependency, so the same code runs offline and in a Pyodide browser lane; the DTW matrix has an optional compiled backend that is parity-tested against the NumPy path. The catalogue and the conformal calibration serialise to JSON with an exact round-trip, so an offline-baked catalogue is replayed client-side. The package is seeded and deterministic, unit-tested (preprocessing, DTW parity, PAM, catalogue round-trip, the conformal coverage-and-OOD test), and published to PyPI as `pygeotypes` under the MIT license. Source and the reproducible validation of this note are at https://github.com/fsantibanezleal/CAOS_GeoTypes.

VII. Limitations

The scope is honest. The shape-catalogue methodology is that of Kamel Targhi *et al.* [1]; this note contributes the reusable implementation and the conformal layer, not the clustering method. The conformal guarantee is finite-sample and class-conditional but rests on exchangeability of calibration and test curves; a covariate shift between the ensemble a catalogue was built on and the signals it is later applied to would weaken it, and class-conditional calibration needs enough curves per type ($n_g \gtrsim 1/\alpha - 1$ for an empty set to be reachable at level α). The validation here is on synthetic dual-porosity transients where ground truth is controlled; on field data the ground-truth type is itself uncertain, and the coverage statement is then relative to the catalogue's own metric. The out-of-catalogue flag detects shapes far from every medoid under DTW, not every kind of anomaly. None of these is hidden: the package's validation document and tests state what is and is not guaranteed.

VIII. Conclusion

`pygeotypes` makes the GeoType shape-catalogue methodology of Kamel Targhi *et al.* reusable and decision-ready: a dependency-light, browser-safe implementation of the full pipeline, plus a class-conditional split-conformal assignment layer that turns catalogue lookup into a guaranteed decision with an honest out-of-catalogue flag. On a reproducible benchmark the coverage guarantee holds across the operating range, the prediction sets stay informative, and alien shapes are flagged. The value is a portable, honestly-validated tool for characterising physical systems by the shape of their response, with the statistical humility to say when a signal does not fit.

Appendix A
Reproduction

The validation is `manuscripts/shape-catalogue-conformal` it builds the catalogue from a seeded Warren-Root ensemble, fits the `ConformalAssigner` on the calibration split, measures marginal and per-class coverage and mean set size across $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$ on the test split, sweeps the out-of-catalogue empty-set rate over the same grid, and writes `data/coverage_validation.json` and the medoid curves. The figure script reads those artifacts; the pipeline schematic is a hand-authored theme-aware drawing. Re-running reproduces every number in Section V and Figs. 2–3 (the run is seeded).

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